Beyond Monolingual Boundaries: Sarcasm Detection on Code-Switched Hinglish Social Media Text

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Motivation

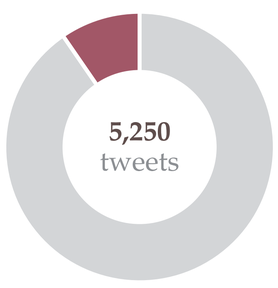
- **Code-switching** (Hindi + English) dominates Indian social media on Twitter and Reddit.
- **Sarcasm** requires pragmatic, figurative, and cultural inference, far beyond sentiment analysis.
- Standard NLP often **breaks** on informal, Romanized Hinglish mixed within a single tweet.
- *"Waah, kya service hai!"* is obvious sarcasm to a bilingual reader, yet many models assign *positive* sentiment.

Research Goals

1. Compare three paradigms under identical conditions.
2. Measure prompt sensitivity in zero-shot LLM use.
3. Evaluate limits of cross-lingual figurative transfer.

Dataset

Hinglish Sarcasm Dataset (Swami et al., 2018)



Split	Size	Sarc.	Rate
Train (80%)	4,200	403	9.6%
Val. (10%)	525	51	9.7%
Test (10%)	525	52	9.9%

References

[1] Aguilar et al. **LinCE: A Benchmark for Language Code-switching Evaluation**. LREC, 2020.

[2] Conneau et al. **Unsupervised Cross-lingual Representation Learning at Scale**. ACL, 2020.

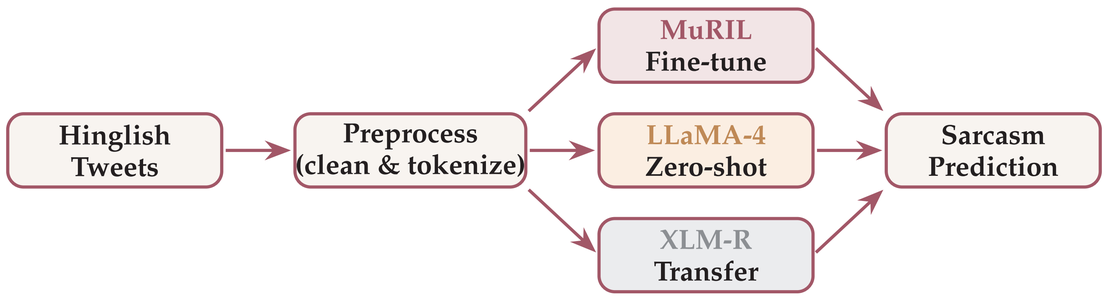
[3] Abu Farha et al. **iSarcasmEval: Intended Sarcasm Detection in English and Arabic**. SemEval, 2022.

[4] Khanuja et al. **MuRIL: Multilingual Representations for Indian Languages**. arXiv:2103.10730, 2021.

[5] Meta AI. **LLaMA 4: Open Foundation and Fine-Tuned Chat Models**. 2025.

[6] Swami et al. **A Corpus of English-Hindi Code-Mixed Tweets for Sarcasm Detection**. arXiv:1805.11869, 2018.

Three-Paradigm Experimental Pipeline



Approach 1: MuRIL

Supervised fine-tuning; class-weighted loss (w=9.33); early stopping on validation Macro F1.

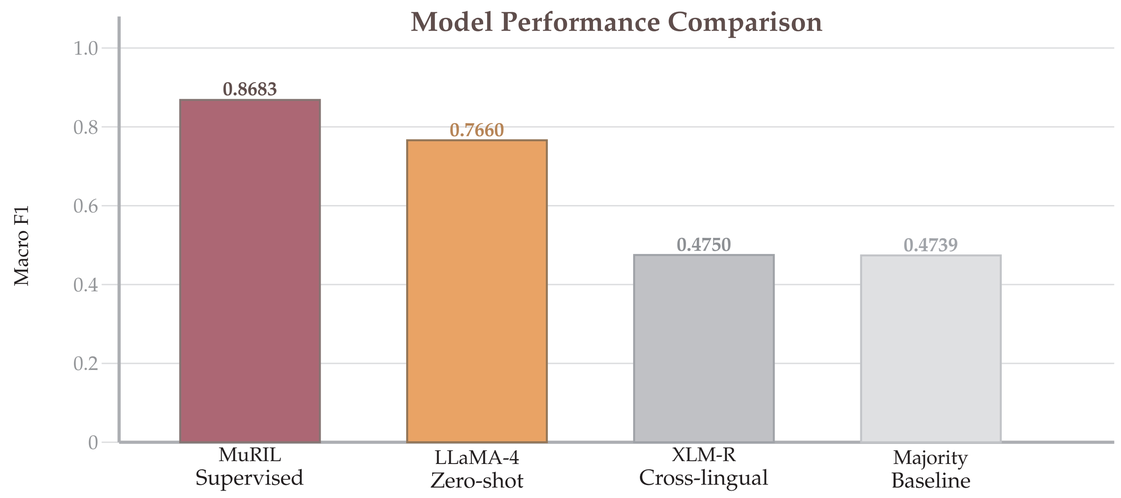
Approach 2: LLaMA-4

Zero-shot prompting via Groq API; calibrated prompt; `temp`= 0 for reproducibility.

Approach 3: XLM-RoBERTa

Trained on English iSarcasmEval; zero-shot transfer to Hinglish after two-phase fine-tuning.

Main Results



Model	Paradigm	Acc.	Macro F1	Sarc. F1
MuRIL	Supervised ft.	**0.9467**	**0.8683**	**0.77**
LLaMA-4 Scout	Zero-shot	0.8971	0.7660	0.59
XLM-RoBERTa	Cross-lingual	0.9048	0.4750	0.00
Majority baseline	Always Non-S.	0.9010	0.4739	0.00

McNemar test: MuRIL vs. LLaMA-4 is statistically significant (χ^2=9.74, p<0.01). XLM-RoBERTa is indistinguishable from the majority baseline.

Confusion Matrices

MuRIL (Supervised)

	P:¬S	P:S
A:¬S	460	15
A:S	13	37

LLaMA-4 Scout (Zero-shot)

	P:¬S	P:S
A:¬S	432	43
A:S	11	39

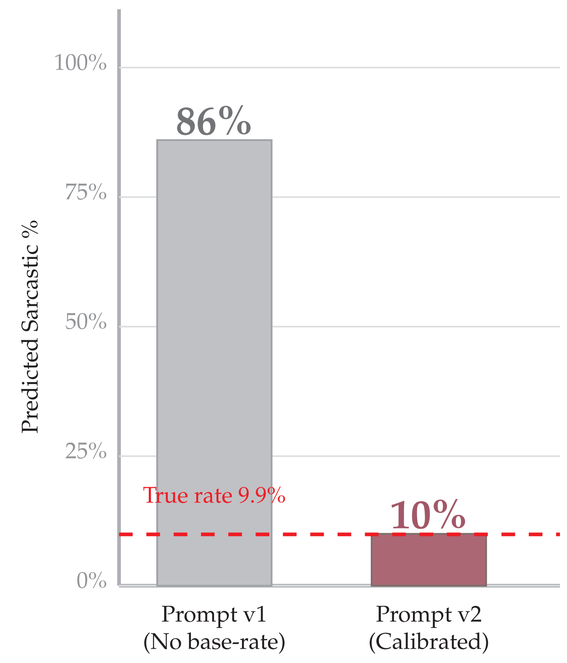
XLM-RoBERTa (Cross-lingual)

	P:¬S	P:S
A:¬S	473	0
A:S	52	0

Interpretation:

- **MuRIL:** Balanced errors (15 FP, 13 FN) reflect genuine sarcasm understanding through in-domain fine-tuning.
- **LLaMA-4:** Recall bias (43 FP vs. 11 FN) shows over-prediction of sarcasm under zero-shot conditions.
- **XLM-RoBERTa: Zero** sarcastic tweets detected; confirms cross-lingual figurative transfer collapses without target-domain adaptation.
- **Precision vs. Recall:** Interestingly, LLaMA-4 detected slightly **more true sarcastic tweets (39 TP) than MuRIL (37 TP)**, but MuRIL's far superior precision ultimately drives its higher F1 score.
- **False Positive Drivers:** Qualitative review of **LLaMA-4's 43 false positives** indicates the model frequently misinterprets highly expressive, non-sarcastic registers (e.g., strong patriotism) as mock-praise.

Prompt Sensitivity



Prompt	Pred. Sarc.	Macro F1
v1 (no base-rate)	≈86%	≈0.27
v2 (calibrated)	≈10%	0.77
True base rate	9.9%	—

One prompt revision = +40 F1 points.

Qualitative Errors

[FP] False Positive — LLaMA-4

"Desh pe garv hai, yeh log sach mein kaam kar rahe hain!"
True: Non-Sarcastic → Pred: Sarcastic
Patriotic register misread as irony.

[FN] False Negative — LLaMA-4

"Interesting decision. Dekh lenge kya hota hai."
True: Sarcastic → Pred: Non-Sarcastic
Understated irony; no mock-praise marker.

[OK] Both Correct — MuRIL & LLaMA-4

"Wah bhai, kya system hai. Form bhar diya, server down."
True: Sarcastic → Pred: Sarcastic
*Explicit mock-praise marker **Wah** signals intent.*

Key Takeaways

- **MuRIL is strongest.** Best Macro F1 (0.8683) with balanced FP/FN errors.
- **LLaMA-4 is competitive but fragile.** Prompt calibration lifts Macro F1 from about 0.27 to 0.766.
- **XLM-R transfer failed.** It collapses to majority-class behavior and misses all sarcastic tweets.
- **Cultural pragmatics are hard to transfer.** Cross-lingual models need explicit domain adaptation or few-shot examples to capture subtle Hinglish irony.